Name: _____ **Unit 1:** Algebra Basics

Date: _____ Bell: _____ **Homework 10:** One & Two-Step Equations

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $y - 17 = -37$

$+17 \quad +17$

$y = -20$

2. $-12 + k = 18$

$+12 \quad +12$

$k = 30$

3. $5m = -20$

$\frac{5m}{5} = \frac{-20}{5}$

$m = -4$

4. $\frac{x}{9} = 3$ $\cdot 9$

$x = 27$

5. $\frac{4}{5}z = 16$ $\cdot \frac{5}{4}$

$z = 20$

6. $3p - 4 = 11$

$+4 \quad +4$

$\frac{3p}{3} = \frac{15}{3}$

$p = 5$

7. $-5a + 7 = -38$

$-7 \quad -7$

$\frac{-5a}{-5} = \frac{-45}{-5}$

$a = 9$

8. $31 - 12n = 211$

$-31 \quad -31$

$\frac{-12n}{-12} = \frac{180}{-12}$

$n = -15$

9. $\frac{x}{3} - 15 = -7$

$+15 \quad +15$

$3 \cdot \frac{x}{3} = 8 \cdot 3$

$x = 24$

10. $\frac{f}{-4} + 11 = 5$

$-11 \quad -11$

$-4 \cdot \frac{f}{-4} = -6 \cdot -4$

$f = 24$

11. $\frac{4}{3}v + 25 = 33$

$-25 \quad -25$

$\frac{3}{4} \cdot \frac{4}{3}v = 8 \cdot \frac{3}{4}$

$v = 6$

12. $\frac{5}{8}c + 13 = 18$

$-13 \quad -13$

$\frac{8}{5} \cdot \frac{5}{8}c = 5 \cdot \frac{8}{5}$

$c = 8$

13. $-\frac{2}{3}w + 53 = 59$

$-53 \quad -53$

$-\frac{3}{2} \cdot -\frac{2}{3}w = 6 \cdot -\frac{3}{2}$

$w = -9$

14. $11 = \frac{4}{5}h - 1$

$+1 \quad +1$

$\frac{5}{4} \cdot 12 = \frac{4}{5}h \cdot \frac{5}{4}$

$15 = h$

15. $-\frac{3}{8}r + 41 = -22$

$-41 \quad -41$

$-\frac{8}{3} \cdot -\frac{3}{8}r = -63 \cdot -\frac{8}{3}$

$r = 168$

16. $34 = 28 - \frac{2}{5}y$

$-28 \quad -28$

$-\frac{5}{2} \cdot 6 = -\frac{2}{5}y \cdot -\frac{5}{2}$

$-15 = y$

17. $\frac{n+4}{-2} = -9 \cdot -2$

$n + 4 = 18$

$-4 \quad -4$

$n = 14$

18. $-5 = \frac{x-1}{8} \cdot 8$

$-40 = x - 1$

$+1 \quad +1$

$-39 = x$

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